

1) Open Windows Defender Firewall settings on your computer and create a new outbound rule to block all traffic from the Spotify desktop app.

Step 1: Open Windows Defender Firewall

- Press Windows + R.
- Type wf.msc and press Enter.
- The Windows Defender Firewall with Advanced Security window will open.

Step 2: Open Outbound Rules

- Click Outbound Rules in the left panel.
- Click New Rule... in the right panel.

Step 3: Select Program Rule

- Select Program and click Next.
- Select This program path.
- Browse to the Spotify desktop application's executable file.

Step 4: Block the Connection

- Select Block the connection and click Next.
- Keep Domain, Private, and Public selected.
- Click Next.

Step 5: Name the Rule

- Enter the name Block Spotify Traffic.
- Click Finish.

Verification

- Confirm the rule appears under Outbound Rules.
- Open Spotify and verify that its network traffic is blocked.

Screenshot to Include

- Outbound rule and its Properties window showing the program path and Block the connection action.

2) Set up an inbound rule in Windows Defender Firewall to allow incoming connections only for Google Chrome on port 443 (HTTPS).

Step 1: Open Inbound Rules

- Click Inbound Rules in Windows Defender Firewall with Advanced Security.
- Click New Rule...

Step 2: Select Port

- Select Port and click Next.
- Select TCP.
- Select Specific local ports and enter 443.
- Click Next.

Step 3: Allow the Connection

- Select Allow the connection.
- Click Next.
- Keep the required network profiles selected.

Step 4: Name the Rule

- Enter the name Allow Chrome HTTPS 443.
- Click Finish.

Important Note

- The task requires the rule to be associated with Google Chrome.
- If a program-specific rule is required, specify the Chrome executable.
- Use TCP port 443 for HTTPS traffic.

Screenshot to Include

- Rule creation window showing TCP and port 443, and the final rule under Inbound Rules.

3) Document, step-by-step, how you would block all traffic from a specific IP address (for example, 123.45.67.89) using Windows Defender Firewall.

Step 1: Open Windows Defender Firewall

- Press Windows + R.
- Type wf.msc and press Enter.

Step 2: Create a New Rule

- Choose Inbound Rules or Outbound Rules depending on the traffic direction.
- Click New Rule...

Step 3: Select a Custom Rule

- Select Custom and click Next.
- Configure the required traffic direction and program/service scope.

Step 4: Use the Scope Tab

- Go to the Scope section.
- Under Which remote IP addresses does this rule apply?, select These IP addresses.
- Click Add... and enter 123.45.67.89.
- Confirm the address and continue.

Step 5: Block the Connection

- Select Block the connection.
- Keep the required network profiles selected.
- Name the rule Block 123.45.67.89.
- Click Finish.

Verification

- Open the rule Properties and verify that 123.45.67.89 appears under Scope.
- The traffic matching the rule should be blocked.

Screenshot to Include

- Scope tab showing 123.45.67.89 and the final blocking rule.

4) Compare hardware firewalls (like those used in large offices) and software firewalls (like Windows Defender) by listing 3 key differences and a use case for each.

Comparison

- 1. Hardware firewall: A dedicated physical network device. Software firewall: A firewall program running on a computer or operating system.
- 2. Hardware firewall: Usually protects multiple devices or an entire network. Software firewall: Usually controls traffic to and from one device.
- 3. Hardware firewall: Commonly placed at the network boundary. Software firewall: Commonly installed directly on endpoints such as PCs and servers.

Use Cases

- Hardware firewall use case: A large office uses a dedicated firewall at the network gateway to control Internet traffic for many employees and devices.
- Software firewall use case: Windows Defender Firewall on a Windows PC controls inbound and outbound connections for applications and services on that PC.

Comparison Summary

- Hardware Firewall — Physical device — Protects a network — Office/enterprise gateway.
- Software Firewall — Software on an endpoint — Protects an individual system — Windows PC.